

Rishi Raj

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GitHub | LinkedIn | Portfolio

SUMMARY

- Final-year B.E. CSE (AI & ML) student with hands-on experience building full-stack web applications and analyzing data sets. Proficient in problem solving and engineering creative ideas. Dedicated to continuous learning and upskilling, eager to apply emerging technologies to drive scalable, data-driven solutions.

PROGRAMMING SKILLS

- **Languages:** Python, C++, C, HTML, JavaScript
- **Concepts:** Data Structures, Algorithms, Machine Learning
- **Frameworks:** ReactJS, Flask, PyTorch, Spring Boot, Angular
- **Developer Tools:** Git, Github, Cursor
- **Soft Skills:** Problem Solving, Communication

PROJECTS

- **AI-Powered Skin Disease Detection System | *Python, PyTorch, Flask, Git* | GitHub:**
 - Built a PyTorch deep learning model to classify 14 different skin diseases from image data.
 - Developed a lightweight Flask REST API to serve real-time predictions from the trained model.
 - Optimized the inference pipeline for fast local deployment, achieving ~50ms response time per image.
 - Used Git for version control to track model iterations and manage the codebase throughout development.
- **LocalShare – AirDrop for Windows & Linux | *Electron, React, TypeScript, Node.js* | GitHub:**
 - Built a cross-platform local-network file and link-sharing desktop app using Electron with mDNS peer discovery via `bonjour-service`.
 - Enabled sharing of both files and web links (e.g., YouTube URLs) between peers on the same network, in addition to standard file transfer.
 - Developed an Express-based file streaming API with HTTP 206 range support for instant media preview without full download.
 - Designed a strongly-typed IPC bridge (contextBridge) connecting the React/TypeScript UI to the Node.js backend, with a Tailwind CSS v4 interface.
- **High-Performance File Compression Tool | *C++, Data Structures, Algorithms*:**
 - Created a fast file compression tool in C++ using the Huffman Coding algorithm.
 - Built custom data structures (min-heaps, hash maps) from scratch for efficiency.
 - Achieved ~40% average reduction in file sizes while maintaining data integrity.

EDUCATION

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|---|---------------------------------|
| • RNS Institute of Technology
<i>B.E. in Artificial Intelligence & Machine Learning</i> | Bengaluru, India
2023 – 2027 |
| • Kendriya Vidyalaya – DGQA
<i>12th CBSE; GPA: 80/100</i> | Chennai, India
2023 |
| • Kendriya Vidyalaya – DGQA
<i>10th CBSE; GPA: 94.60/100</i> | Chennai, India
2021 |